

Elif Nur Tekay

Junior Full-Stack Software Engineer | AI-Assisted Product Development

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Professional Summary

Software Engineering graduate with hands-on experience in full-stack web development, artificial intelligence, computer vision, multimodal analysis, and reinforcement learning. Built applications using Python, Django, Node.js, React, SQL/NoSQL databases, WebSocket, and WebRTC; also integrated AI models into working web interfaces. Experienced in collaborative research, technical reporting, model development, debugging, and product demonstrations. Awarded second place among 800 teams in Turkcell's technology competition, first place in Ankara University's AI and Data Science category, and the INFUS 2026 Best Student Paper Award.

Education

Malatya Turgut Özal University

B.Sc. in Software Engineering; GPA: 3.37/4.00

Ankara University

Software Engineering, Special Student Status

Malatya, Türkiye

2022 – 2026

Ankara, Türkiye

2023 – 2025

Experience

ANLAM-NET Research LAB, Ankara University

Research Assistant

Ankara, Türkiye

Aug. 2024 – Present

- Contribute to research in artificial intelligence, natural language processing, and multimodal media analysis, including data preprocessing, model development, experimental evaluation, and technical reporting.
- Support academic publications, project applications, competition deliverables, and the integration of video and brand memorability models into a working web interface.

TÜBİTAK SAGE

Artificial Intelligence Applications Intern

Ankara, Türkiye

Jul. 2025 – Aug. 2025

- Fine-tuned and evaluated super-resolution models to enhance details in large-scale satellite imagery.
- Combined processed image patches through stitching and contributed to model comparison and optimization.

Ankara University Project Support Office

Part-Time Student Employee

Ankara, Türkiye

Mar. 2025 – Jun. 2025

- Managed website content and updates, and supported documentation related to academic project processes.

SA Innovation

Intern

Ankara, Türkiye

Jul. 2024 – Aug. 2024

- Cleaned, normalized, and labeled audio and text data for an AI-based sentiment analysis project.

Technical Skills

Programming: Python, Java, C/C++, JavaScript, PHP, SQL, HTML, CSS

Full-Stack: Django, Flask, Node.js, React, Bootstrap, WebSocket, WebRTC

Databases: PostgreSQL, MySQL, MongoDB, SQLite3

AI/ML: Machine Learning, Deep Learning, Computer Vision, Image Processing, Reinforcement Learning, XAI, GANs, Super-Resolution

Tools: Git, GitHub, Docker, Postman, Linux, Google Colab, Visual Studio, VS Code, Claude, OpenAI Codex, ChatGPT

Language: English (B1; academic writing and international presentation experience), Turkish (native)

Selected AI and Engineering Projects

MemoVision: Video and Brand Memorability Analysis Platform

2025 – 2026

TÜBİTAK 2209-A / Turkcell Competition

- Developed CLIP-based multimodal approaches combining visual, temporal, textual, and metadata features for video and brand memorability prediction across VideoMem, Memento10K, and VIDEM; also compared MLP and Vision Transformer models for memorability classification in an IEEE UBMK 2025 study.
- Connected trained prediction models to a working web application for content-level and brand-level analysis; the project placed second among 800 teams in Turkcell's competition.

Aerodynamic Shape Optimisation Using Reinforcement Learning

2025 – 2026

TUSAŞ LIFT UP Capstone Project – Completed

- Contributed to a CST-based airfoil shape optimization framework using PPO, TD3, and SAC, supported by hybrid surrogate models and XFOIL/NeuralFoil validation under aerodynamic and geometric constraints.
- Applied SHAP-based explainable AI methods to trained reinforcement learning policies to analyze how airfoil geometry and operating parameters influenced the agents' optimization decisions.
- Integrated optimization results and explainability outputs into a working web interface; the project won first place in Ankara University's AI and Data Science category and produced the INFUS 2026 Best Student Paper Award-winning publication.

AI-Based Perception, Tracking, and Position Estimation for Autonomous Aerial Vehicles

2026 – Present

TEKNOFEST Aviation AI Competition / Ankara University BAP – Ongoing

- Contribute to the three competition tasks: UAV and helicopter object detection and tracking, camera-based position estimation, and reference object detection.

- Contribute to Python-based model integration, inference and evaluation pipelines, performance analysis, output validation, and collaborative development using Git and GitHub.

Predicting Movie and Commercial Memorability

2025

MediaEval 2025 Benchmark

- Compared classical machine-learning and deep-learning models using multiple feature combinations and fusion strategies on the VIDEM dataset; results were published in the MediaEval 2025 Workshop Proceedings.

Full-Stack Web Projects

Project Management Application

Django, MongoDB, React, CSS

- Built a web application with task assignment, progress tracking, team management, and Gantt chart visualizations.

Student Attendance Application

Python, Django, SQLite3

- Developed a web-based attendance management system using Django and SQLite3.

Real-Time Messaging and Calling Application

Node.js, WebSocket, WebRTC, MySQL

- Implemented real-time messaging, voice calling, and video calling in a browser-based communication application.

AjandaX - Event and Appointment Management System

Django, MongoDB, HTML, CSS, Bootstrap, JavaScript

- Created a calendar-based application for adding, editing, viewing, and managing events and appointments with reminders.

Publications

Comparative Analysis of MLP and Vision Transformers for Video Memorability Classification

2025

IEEE UBMK 2025

- E. N. Tekay, I. A. Isleyen, B. Gungor, M. Karakus, and R. S. Kiziltepe. DOI: 10.1109/UBMK67458.2025.11207010.

Multimodal Feature Fusion for Video and Brand Memorability

2025

MediaEval 2025 Workshop Proceedings

- E. N. Tekay, I. A. Isleyen, M. Karakus, and R. S. Kiziltepe. Paper link.

Reinforcement Learning-Based Airfoil Shape Optimization Using Hybrid Surrogate Models

2026

INFUS 2026 - Presented; Best Student Paper Award

- Conference paper produced from the completed TUSAŞ LIFT UP capstone project and presented at INFUS 2026.

A CLIP-Based Visual-Semantic Embedding Approach for Commercial Memorability Prediction

2026

MediaEval 2026 Workshop Proceedings

- E. N. Tekay, I. A. Isleyen, R. S. Kiziltepe, and M. Karakus. Paper link.

Awards

Best Student Paper Award

Jul. 2026

INFUS 2026

First Place - Artificial Intelligence and Data Science Category

May 2026

Ankara University IT Project Market and Career Days

Second Place - MemoVision

May 2026

Turkcell Tomorrow's Technology Leaders Competition; 824 teams

Selected Certificates

Data Analysis School 2025–2026 – Artificial Intelligence and Machine Learning Module, Certificate of Achievement (Council of Higher Education – YÖK) | Web Development with Django (BTK Academy) | C++ Programming (BTK Academy) | Java Programming (BTK Academy) | Deep Learning Bootcamp | Machine Learning Bootcamps (Akbank & Aygaz) | Python Programming 401 | White Hat Hacker (Siber Vatan) | Introduction to Cybersecurity | Introduction to Linux and Networks (Cisco)

Reference

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